Chidiebere Emmanuel Udegbe

Mechatronics Engineer · Robotics, IoT & Embedded Systems

audegbe2003@gmail.com | github.com/dist0rted777 | dist0rted777.github.io | Nigeria — open to remote / relocation

Profile

Mechatronics engineer who designs and builds working robots and autonomous systems end-to-end — from mechanical CAD and electrical schematics to embedded firmware and a tested build. Comfortable across hardware, software and applied AI. Also a music producer and video editor (*DIST0RTED*). Strong at turning an idea into a physical, functioning prototype.

Selected Projects

Wi-Fi Teach-and-Repeat Mobile Robot (Final-year project) — ESP32 · IMU + odometry · L298N
A tank-tread robot that records a manually driven path and autonomously repeats it, using low-cost inertial and odometric sensing. Full CAD, electrical schematics and working build.

Automatic Solar-Powered Maize Sheller — Arduino · SolidWorks · Solar + MPPT · KiCad
Solar-powered machine that shells maize automatically — mechanical CAD, drive-train design, control schematics and a field-built prototype.

IoT Prepaid Energy Meter & Load Management — ESP32 · ACS712 · Blynk / ThingSpeak
Prepaid energy meter with automated load management and a web dashboard for real-time monitoring and control.

IoT Hybrid Access Control System — ESP32 · RFID (RC522) · Keypad · Mobile app
Hybrid RFID + keypad + app access-control system for remote property management, with mobile and web interfaces.

Robotic Assistant for Hospital Supply Delivery — ESP32 · DC drive · Onboard camera
Autonomous cart robot for ferrying supplies through hospital corridors — CAD chassis, power circuit and built prototype.

Further projects: autonomous line-guided warehouse delivery robot, dual-purpose solar-powered scarecrow, smart-house electrical design. Full portfolio at dist0rted777.github.io.

Technical Skills

CAD / Mech	SolidWorks, Fusion 360, Autodesk Inventor; 3D modeling, prototyping, 3D printing
Electronics	KiCad, EasyEDA; schematic & PCB design, sensors & motor control, solar / MPPT power
Embedded	ESP32, Arduino (C/C++); microcontrollers, PWM motor drive, PID control
AI / Autonomy	Computer vision, sensor fusion, path planning, ML & model training (Python), LLM / agent building
IoT	Blynk, ThingSpeak; Wi-Fi dashboards, remote monitoring & control
Creative	Premiere Pro, DaVinci Resolve, CapCut (video); FL Studio (music production)
Tools	Python, Git / GitHub

Education

B.Eng, Mechatronics Engineering — Afe Babalola University, Ado-Ekiti, Nigeria
College of Engineering — Department of Mechanical & Mechatronics Engineering